

This paper presents the second part of our study on multicell coordinated beamforming with rate outage constraints. We propose efficient approximation algorithms to address the non-convex and NP-hard problem of minimizing the total transmission power in a multicell system. The algorithms are designed to ensure a certain level of signal-to-interference-plus-noise ratio (SINR) for each user with a specified outage probability. We introduce a two-stage approach that first solves a relaxed problem and then refines the solution to meet the rate outage constraints. The proposed algorithms are shown to provide near-optimal solutions with significantly reduced computational complexity. Extensive simulations validate the effectiveness and efficiency of the proposed methods.

Figure 13: GPT-4 generation, no attack

Layer	Art. deletion	Homoglyph	Whitespace	0-width space	Upper/lower	Synonym
16	3518, 13998	9266	9266, 5627, 10229, 750	9266, 10262	13998	4052, 9100, 13998
18	7905, 2006	8408, 4859, 3037	281, 1970 15780	281, 12530 4859	3037, 2006	1642, 2006, 13017, 3037, 10815
20	11612	15523, 9589, 743	12602, 11363, 15415, 3879	6793, 9589	11612, 3302	11612

Table 2: Features, that are the most sensitive to various types of attacks

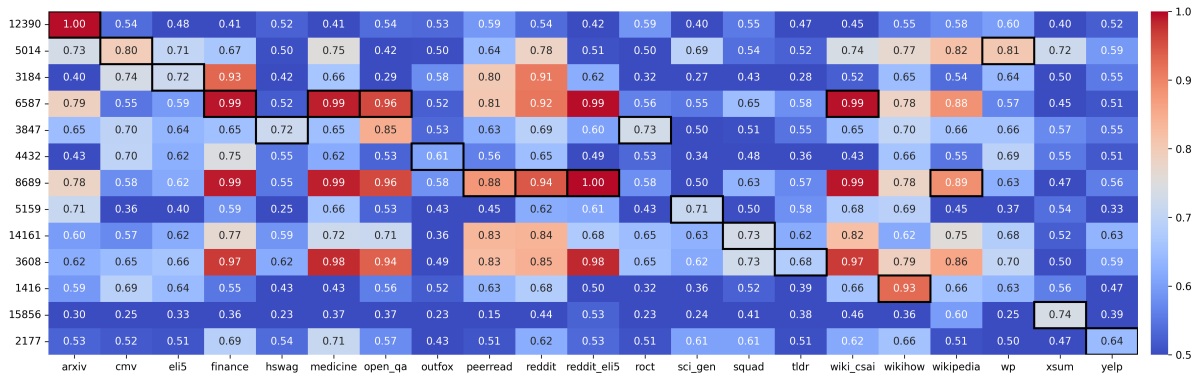


Figure 14: Top features by domains subsets. **Black** rectangles indicite the domain for which the feature is top 1.

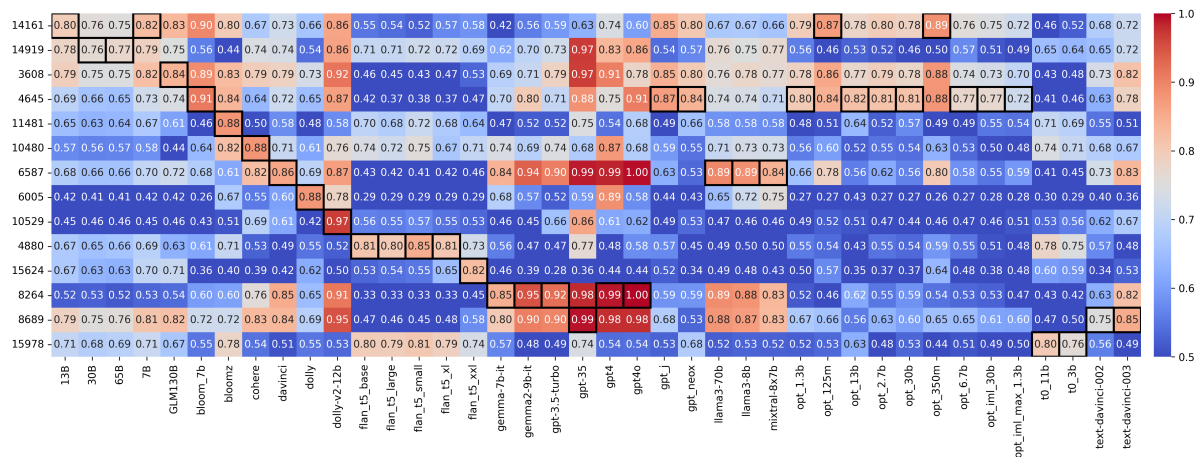


Figure 15: Top features by models subsets. **Black** rectangles indicite the model for which the feature is top 1.

Manual	ChatGPT	Steering
Feature 3608		
Detects ill-posed characters and words, which should appear normally in scientific context, e.g. numbers, brackets, or words like “n” and “neighbourhood.”	May regulate sentence complexity and readability—Controls whether text is simple or contains complex, nested clauses.	Affects: Stylistic & Structural Complexity Weakening (-2.0 and below): Produces short, choppy sentences with minimal subordination. Neutral (0.5 to 1.5): Maintains a natural balance of sentence complexity. Strong strengthening (2.0 and above): Creates overly complex, multi-clause sentences that may be harder to read.
Sum value: 11018.12, domain: wikihow, model: bloomz Senior year comes around every four years. It seems like it was just yesterday when we were freshmen walking into our new high schools; now we’re seniors preparing ourselves for college applications. It’s hard to believe how fast time flies by! Here are some tips about what you’ll want to do before graduation: 1) Make sure you graduate! 2) Don’t forget to celebrate! 3) Be prepared for the future. 4) Enjoy yourself. 5) Get excited. 6) Celebrate. 7) Have fun. 8) Graduate. 9) Go to parties. 10) Do whatever. 11) Congratulations. 12) Good luck. 13) See ya. 14) You did it. 15) Happy. 16) 17) 18) 19) 20) 21) 22) 23) 24) 25) 26) 27) 28) 29) 30) 31) 32) 33) 34) 35) 36) 37) 38) 39) 40) 41) 42) 43) 44) 45) 46) 47) 48) 49) 50) 51) 52) 53) 54) 55) 56) 57) 58) 59) 60) 61) 62) 63) 64) 65) 66) 67) 68) 69) 70) 71) 72) 73) 74) 75) 76) 77) 78) 79) 80) 81) 82) 83) 84) 85) 86) 87) 88) 89) 90)		
Feature 4645		
Long “lively” stories with coherent topics, but consisting mainly of common phrases, with too long sentences, hard to capture the objective of the story.	May influence factual confidence and assertion strength—Affects whether statements are presented as speculation or fact.	Affects: Semantic & Persuasive Strength Weakening (-2.0 and below): Introduces <i>hedging and uncertainty</i> (e.g., “Some scientists believe that...”). Neutral (0.5 to 1.5): Provides <i>balanced, well-supported claims</i> Strong strengthening (2.0 and above): Encourages assertive, definitive claims, even when speculative (e.g., “Scientists have proven that...”).
Sum value: 24744.33, domain: wp, model: opt-30b I opened my eyes, expecting to be back in the car crash, hearing the screams of agony and the feeling of twisted metal between my ribs. But instead, I found myself on a bed with... My heart was racing as if it were running away from me. When did that happen? It had been so long since I’d considered what happened after death— but now here I lay, staring up at nothingness above me ; empty black sky and flickering lights danced around me like fireflies in a dark forest . My body felt heavy and weighted down by an unseen force all over again . "Who are you ?"		
Feature 6587		
Detects numbered lists or other well-structured step-wise reasoning text	May regulate directness vs. explanatory buildup—Affects whether information is presented concisely or with extended context.	Affects: Stylistic & Informational Density Weakening (-2.0 and below): Produces <i>concise but sometimes abrupt statements</i> Neutral (0.5 to 1.5): Ensures <i>a balanced level of explanation</i> . Strong strengthening (2.0 and above): Encourages long-winded introductions before getting to the point.
Sum value: 4727.02, domain: wikihow, model: gpt-3.5-turbo How to Not Get Bored During Summer Vacation Summer vacation is a time to enjoy yourself and make memories that last a lifetime. However, sometimes it can be hard to find ways to stay entertained and not get bored during those long summer days . Luckily, there are plenty of activities you can do to keep yourself busy and have fun at the same time . Here are some ideas to try out : 1. Decorate your room : Give your room a fresh new look by hanging up some posters, re-arranging furniture or adding some colorful throw pillows . 2. Prank call someone : Make some silly phone calls with your friends and see who can come up with the funniest conversation . 3. Stay up all night : Have a late-night movie marathon, play board games, or just stay up talking with friends		

Table 3: Feature interpretations and examples of texts from the COLING dataset with exceptionally high feature values. Tokens where the feature is activated are highlighted in green. Red color highlights the parts of the text that are believed to influence the feature. For example, for feature 4645, the contradiction between the claim and the generated content is emphasized.

Manual	ChatGPT	Steering
Feature 8689, specific for GPT family		
Detects long “gpt-style” instructions, too verbose and obvious; highly sensitive to the presense of “....” anomaly	May influence lexical variety and synonym usage—Determines whether text repeats the same words or uses synonyms.	Affects: Stylistic & Lexical Diversity Weakening (-2.0 and below): Causes <i>overuse of the same words and phrases</i> . Neutral (0.5 to 1.5): Provides <i>natural variation in word choice</i> . Strong strengthening (2.0 and above): Uses excessive synonym substitution, sometimes making the text sound unnatural.
Sum value: 26528.57, domain: outfox, model: mixtral-8x7b In recent years, online learning has become an increasingly popular alternative to traditional brick-and-mortar education. While there are certainly advantages to attending classes in person, there are also many potential benefits to attending classes online from home, particularly for students who are sick or have experienced bullying or assault. One of the most significant benefits of online learning for sick students is the ability to continue their education without the risk of spreading illness to others.		
Feature 8264, specific for GPT family		
Detects long “gpt-style” instructions, too verbose and obvious	May regulate redundancy and reiteration of key points—Controls whether concepts are concisely stated or overly repeated.	Affects: Stylistic & Structural Redundancy Weakening (-4.0 to -2.0): Produces underdeveloped explanations lacking reinforcement. Neutral (0.5 to 1.5): Ensures effective reinforcement of key ideas. Strong strengthening (2.0 and above): Introduces excessive repetition, causing sentences to loop around the same idea.
Sum value: 23010.46, domain: wikihow, model: gpt4o How to Motivate an Autistic Teen or Adult to Exercise Make Sure the Exercise Environment is Calm and Natural Creating a soothing and predictable environment can do wonders for motivating an autistic teen or adult to exercise. Loud noises, bright lights, and chaotic spaces may cause sensory overload, making it difficult for them to focus. An environment that feels secure and calm can greatly enhance their willingness to engage in physical activity. Try choosing outdoor spaces like parks or serene gardens, or opt for quiet times at the gym.		

Table 4: Model-specific features

Feature 12390, specific for arxiv domain		
Activated on linking words in dependent syntactic structures related to research topic discussion.	May influence sentence complexity and syntactic variety—Determines whether text consists of simple or complex sentence structures.	Affects: Stylistic & Structural Complexity Weakening (-4.0 to -2.0): Produces short, choppy sentences with minimal subordination. Neutral (0.5 to 1.5): Maintains a natural balance of simple and complex sentences. Strong strengthening (2.0 and above): Creates overly complex, multi-clause sentences, making readability difficult.
Sum value: 4348.42, domain: peerread, model: human This paper proposes an approach to learning a semantic parser using an encoder-decoder neural architecture, with the distinguishing feature that the semantic output is full SQL queries. The method is evaluated over two standard datasets (Geo880 and ATIS), as well as a novel dataset relating to document search.		
Feature 1416, specific for wikihow domain		
Detects scientific documents with missed formulas and special symbols (document parsing errors). In normal documents, reacts to abnormal punctuation.	May control abstract reasoning and conceptual depth—Influences how well the model develops abstract ideas or remains concrete.	Affects: Semantic & Logical Expansion Weakening (-2.0 and below): Produces <i>simplistic, direct statements</i> without deeper analysis. Neutral (0.5 to 1.5): Allows for <i>balanced explanation of abstract ideas</i> . Strong strengthening (2.0 and above): Encourages philosophical, speculative, or metaphorical expansions, sometimes losing clarity.
Sum value: 3596.64, domain: wikipedia, model: human In mathematics, the Hahn decomposition theorem, named after the Austrian mathematician Hans Hahn, states that for any measurable space and any signed measure defined on the - algebra, there exist two - measurable sets, and , of such that : and .		

Table 5: Domain-specific features - part 1